



Vibration-Guided Neural Separation of Overlapping Bird Vocalizations

Weak Supervision via Simulated On-Body Vibrations

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The authors declare that they are the sole authors of this thesis and that they have not used any sources other than those listed in the bibliography and identified as references. They further declare that they have not submitted this thesis at any other institution to obtain a degree.

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Abstract

Passive acoustic monitoring generates vast archives of bird recordings, yet automated analysis degrades sharply when multiple species vocalize simultaneously. Existing audio-only separation systems based on Conv-TasNet struggle particularly under equal-energy overlap conditions, where both sources contribute identical power to the mixture and energy-based masking provides no discriminative signal. This thesis investigates whether computationally simulated, accelerometer-inspired vibration signals can improve neural audio source separation of overlapping bird vocalizations without requiring physical sensors. A vibration signal is synthesised from the mixture audio by extracting a short-time RMS envelope, applying low-pass filtering to retain sub-300 Hz mechanical content, and adding sensor noise and sparse transient artefacts. This simulated signal conditions a Conv-TasNet separator via Feature-wise Linear Modulation at the bottleneck, while an auxiliary per-source vibration supervision loss regularises training. Controlled experiments on synthetic two-source mixtures drawn from a five-species corpus show that the multimodal model outperforms the audio-only baseline by 0.966 dB in SI-SDR improvement on the standard test set, widening to 1.077 dB under equal-energy overlap. Spectral convergence and STFT metrics confirm that spectral fidelity is preserved alongside the energy-domain improvement, supporting the suitability of the separated outputs for downstream bioacoustic analysis.

Keywords: audio source separation, bioacoustics, vibration simulation, multimodal learning, weak supervision

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On a summer morning in a temperate woodland, a passive acoustic recorder captures hours of continuous audio. Within any given ten-second window, three, four, or five species may be vocalizing simultaneously. The recorder captures everything faithfully; the analysis pipeline that follows does not. Automatic species recognisers trained on clean, single-species recordings degrade substantially when their input is a superposition of overlapping vocalizations, and the degradation worsens precisely when ecological interest is highest — during dawn choruses, breeding season peaks, and high-density habitats where biodiversity is greatest. This tension between the scale at which passive acoustic monitoring can operate and the quality of signal that downstream analysis requires is the central problem this thesis addresses.

1.1 Background

Passive acoustic monitoring (PAM) has emerged over the past two decades as the dominant scalable method for large-area biodiversity assessment. Autonomous recording units can operate continuously for weeks or months, accumulating data at a spatial and temporal resolution that is impossible to match with human observers [18]. The resulting archives are analysed using automated recognisers, of which BirdNET [9] is among the most widely deployed, achieving high accuracy on clean single-species recordings across hundreds of species. However, the fundamental assumption underlying these systems — that the target species is the dominant or only source in the recording — fails routinely in natural soundscapes.

The computational problem of recovering individual source signals from a single-channel mixture is known as *audio source separation* or, in the speech domain, the *cocktail party problem* [21]. Deep learning approaches to this problem have matured rapidly. Permutation-invariant training (PIT) [24] resolved the label-assignment ambiguity that had stalled earlier supervised methods. Conv-TasNet [14] subsequently demonstrated that end-to-end waveform-domain separation using a temporal convolutional network could surpass classical time-frequency masking in both speech quality and computational efficiency. These advances have been thoroughly characterised in speech separation benchmarks; their transfer to bioacoustic signals involving non-human, highly non-stationary vocalizations from diverse species is considerably less studied.

The difficulty of separating overlapping bird vocalizations relative to speech arises from several compounding factors. Species-specific vocalizations exhibit a far wider range of temporal and spectral structure than human speech, varying from broadband

impulse trains to narrow tonal sweeps. Unlike in the speech case, no strong linguistic or prosodic prior constrains the signal. Furthermore, the equal-energy, high-overlap condition ($\text{SNR} \approx 0$ dB) is common in chorusing contexts and constitutes the hardest possible case for energy-based or mask-based separation methods.

A distinct sensing modality that partially circumvents the acoustic mixing problem is on-body vibration measurement. Contact microphones and accelerometers mounted on or near a vocalizing animal capture mechanical vibrations transmitted through the body or substrate, providing a signal dominated by the vocalizations of the attached animal and largely insensitive to airborne interference from other species [23]. In controlled laboratory settings, such sensors have been used to record individual bird vocalizations with high source isolation. The limitation is obvious: attaching sensors to wild animals at population scale is operationally infeasible, and PAM infrastructure does not include contact sensing by design.

This thesis proposes to bridge the gap between the information content of on-body vibration signals and the operational constraints of PAM by *simulating* vibration signals computationally from existing clean audio. The simulation approximates the low-frequency temporal energy envelope that a contact microphone would capture, adds sensor noise and sparse transient artefacts, and produces a signal that encodes the activity timing and relative energy of an individual source without requiring any physical sensor. These simulated signals are used in two complementary roles: as a weak supervision target against which an auxiliary output head of the separator is trained, and as a conditioning input to the separator via Feature-wise Linear Modulation (FiLM) [17], applied to the mixture vibration signal. Formal definitions of all key concepts and metrics are given in Chapter 2.

1.2 Problem Statement and Scope

The central research problem is: *can computationally simulated, accelerometer-inspired vibration signals improve the neural audio source separation of overlapping bird vocalizations, and if so, through which mechanism — conditioning, auxiliary supervision, or their combination?*

The scope of this work is deliberately constrained along three dimensions. First, the separation problem is restricted to two simultaneously vocalizing sources, consistent with the two-output Conv-TasNet architecture; extension to three or more sources is identified as future work. It should be noted that this two-source restriction is not only an architectural simplification but also an ecological one: PAM recordings during dawn choruses routinely contain more than two simultaneously vocalizing species, and the results of this study therefore apply specifically to pairwise-overlap conditions. Multi-source scenarios represent a qualitatively harder problem that is not addressed here. Second, the evaluation dataset is drawn from a single publicly available five-species corpus to ensure reproducibility and to allow controlled manipulation of overlap conditions; generalization to multi-habitat, multi-continent corpora is not claimed. Third, the vibration signals used throughout are simulated, not measured. A single-channel contact microphone mounted on a stationary structure — rather than on the bird — is proposed as the most practically feasible extension to a real sensor, and the mixture vibration signal used for conditioning is consistent with

this design.

The following three research questions operationalize the research problem:

- RQ1** How does conditioning a neural audio source separation model with simulated, accelerometer-inspired vibration signals influence separation performance across different levels of source overlap?
- RQ2** How effective are simulated vibration signals, derived from clean bird audio, as weak supervisory guidance for multimodal source separation relative to an audio-only training objective?
- RQ3** To what extent does vibration-guided separation preserve the temporal and spectral characteristics of individual bird vocalizations in the recovered waveforms?

RQ1 addresses the primary performance claim of the thesis and is answered by comparing Model A (audio-only Conv-TasNet baseline) against Model B (vibration-conditioned MultimodalSep) on both a balanced test set and a dedicated high-overlap subset. RQ2 isolates the contribution of the auxiliary vibration loss by ablating the loss weight λ . RQ3 is addressed through spectral convergence and STFT ℓ_1 metrics computed on the same test splits.

1.3 Thesis Outline

The remainder of this thesis is organised as follows.

Chapter 2 introduces the technical background for this thesis, defining key concepts and mathematical notation used throughout, including the primary evaluation metrics, the conditioning mechanism, and the training paradigm.

Chapter 3 surveys the literature on audio source separation, multimodal conditioning, and bioacoustic analysis, situating the proposed approach within existing work and identifying the specific research gap it addresses.

Chapter 4 describes the experimental design in detail, covering the dataset and recording-level split strategy, the vibration simulation pipeline, the architectures of both models, the loss functions, and the evaluation protocol.

Chapter 5 presents the quantitative results on both test conditions, including metric breakdowns by overlap level, training convergence behaviour, and spectral quality measurements.

Chapter 6 interprets the results in relation to the three research questions, discusses the mechanisms by which vibration conditioning provides benefit, addresses the validity and limitations of the simulation approach, and situates the findings within the broader bioacoustics and multimodal learning literature.

Chapter 7 states the conclusions, summarises the contributions, and proposes directions for future work.

1.3.1 Ethical, Societal, and Sustainability Aspects

Ethical aspects. This study relies exclusively on publicly available bird vocalization datasets and does not involve interaction with live animals, deployment of

physical sensors, or collection of personal data. All source material is used in accordance with its respective open-access terms, ensuring compliance with ethical research standards and reproducibility principles.

Societal impact. Improved automated separation of overlapping bird vocalizations directly supports biodiversity monitoring at scale, enabling more accurate species abundance estimates that inform conservation policy and habitat management decisions. At the same time, increased reliance on automated analysis tools may reduce the incentive to develop human expertise in bioacoustics, and errors in automated outputs — particularly false absences of rare species — carry meaningful ecological risk if not validated by domain experts.

Sustainability aspects. The proposed approach improves the analytical yield of recordings already collected by existing passive acoustic monitoring infrastructure, reducing the need for repeat field surveys and their associated travel and equipment costs. Both models train on a single consumer-grade GPU in a matter of hours, representing a modest computational footprint relative to the ecological value of the data they help analyse. As hardware efficiency improves and renewable-powered computing infrastructure expands, the environmental cost of such experiments is expected to decrease further.

This chapter introduces the technical concepts and mathematical notation used throughout the thesis. It defines the evaluation metrics, modelling primitives, and loss formulations that are referenced by later chapters, particularly Chapter 4. Readers already familiar with audio source separation fundamentals may proceed directly to Chapter 3.

2.1 Fundamental Concepts and Notation

Several terms are used with precise technical meaning throughout this thesis and are defined here to avoid ambiguity.

Scale-Invariant Signal-to-Distortion Ratio (SI-SDR). The primary evaluation metric for source separation quality. Given an estimated waveform $\hat{y}$ and a reference waveform y , both zero-meaned, SI-SDR is defined as:

$$\text{SI-SDR}(\hat{y}, y) = 10 \log_{10} \frac{\|\alpha y\|^2}{\|\hat{y} - \alpha y\|^2}, \quad \alpha = \frac{\hat{y}^\top y}{\|y\|^2} \quad (2.1)$$

where αy is the projection of the estimate onto the reference. SI-SDR is scale-invariant by construction, meaning that a constant gain applied to $\hat{y}$ does not change the score; this makes it more appropriate than raw SDR for systems that do not explicitly control output gain [14]. In experiments, SI-SDR is reported as an improvement over the unprocessed mixture (SI-SDRi):

$$\text{SI-SDRi}(\hat{y}_s, y_s) = \text{SI-SDR}(\hat{y}_s, y_s) - \text{SI-SDR}(y_{\text{mix}}, y_s) \quad (2.2)$$

A positive SI-SDRi indicates improvement over the unprocessed mixture.

Permutation-Invariant Training (PIT). A training strategy for multi-source separation in which the loss is computed under the best assignment of model output slots to ground-truth sources, taken over all possible permutations [24]. For a two-source system, PIT evaluates both assignments ($\hat{y}_1 \leftrightarrow y_1, \hat{y}_2 \leftrightarrow y_2$) and ($\hat{y}_1 \leftrightarrow y_2, \hat{y}_2 \leftrightarrow y_1$) and back-propagates through whichever achieves the higher SI-SDR. This removes the need for a pre-specified source ordering during training.

Feature-wise Linear Modulation (FiLM). A conditioning mechanism in which a secondary input (here, the vibration signal) is used to produce per-channel affine transformation parameters (γ, β) that scale and shift the feature maps of a primary processing stream [17]:

$$\text{FiLM}(h; \gamma, \beta) = \gamma \odot h + \beta \quad (2.3)$$

where h is the feature tensor of the audio processing stream, and γ, β are produced by a learned linear projection of the pooled vibration encoding. FiLM allows the conditioning signal to modulate the behaviour of the main network without requiring the auxiliary pathway to replicate the full depth of the primary network.

Weak supervision. A training paradigm in which auxiliary supervision signals that are cheaper or easier to obtain than the primary labels are used to regularise or guide learning [21]. In this work, the per-source simulated vibration signals constitute weak supervision: they are derived automatically from clean audio without human annotation, and they supervise only an auxiliary output head of the network rather than the primary separation objective.

Recording-level data split. A dataset partitioning strategy in which the train, validation, and test sets are defined at the level of original recordings (individual field recordings or studio sessions), not individual clips derived from them. This is necessary to prevent data leakage: if clips from the same underlying recording appear in both the training and test sets, the model may overfit to recording-specific acoustic conditions (room acoustics, microphone characteristics, background noise profiles) rather than learning generalizable separation cues.

2.2 Model Equations

The following equations describe the core computations of the two models and are referenced throughout Chapter 4.

Encoder. The Conv-TasNet encoder applies a bank of $N = 256$ learned filters of length $L = 16$ samples to the mixture waveform, producing a latent representation $E \in \mathbb{R}^{N \times T'}$:

$$E = \text{ReLU}(W_{\text{enc}} * y_{\text{mix}}) \quad (2.4)$$

Short-time RMS envelope. The vibration simulation pipeline frames the input waveform with size $F = 512$ samples and hop $H = 128$ samples. The RMS of each frame is:

$$r_k = \sqrt{\frac{1}{F} \sum_{n=0}^{F-1} y[kH + n]^2} \quad (2.5)$$

FiLM conditioning parameters. The vibration embedding $\bar{v}$ is projected through a two-layer MLP to produce per-channel affine parameters:

$$[\gamma, \beta] = \text{MLP}(\bar{v}), \quad \gamma, \beta \in \mathbb{R}^B \quad (2.6)$$

These are applied as a feature-wise transform to the bottleneck representation $H \in \mathbb{R}^{B \times T'}$ before the DSConv stack:

$$\tilde{H}_{b,t} = \gamma_b \cdot H_{b,t} + \beta_b \quad (2.7)$$

2.3 Loss Functions

Primary audio separation loss. Both models are trained with a permutation-invariant SI-SDR loss. The best source-to-output assignment π^* is found over all permutations $\mathcal{S}_2$:

$$\mathcal{L}_{\text{audio}} = -\frac{1}{2} \sum_{s=1}^2 \text{SI-SDR}(\hat{y}_s, y_{\pi^*(s)}), \quad \pi^* = \arg \max_{\pi \in \mathcal{S}_2} \sum_s \text{SI-SDR}(\hat{y}_s, y_{\pi(s)}) \quad (2.8)$$

Auxiliary vibration loss. The VibrationHead outputs $\hat{v}_1, \hat{v}_2$ are supervised against the per-source simulated vibration targets using the same permutation π^* :

$$\mathcal{L}_{\text{vib}} = \frac{1}{2} \sum_{s=1}^2 \|\hat{v}_s - v_{\pi^*(s)}\|_2^2 \quad (2.9)$$

Combined loss. The total training loss for Model B is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{audio}} + \lambda \cdot \mathcal{L}_{\text{vib}} \quad (2.10)$$

where λ is a weighting factor that balances the two objectives. For Model A, $\lambda = 0$ and no VibrationHead is instantiated.

2.4 Spectral Quality Metrics

In addition to SI-SDRi, two spectral quality metrics are used in this thesis to evaluate whether separation preserves the fine-grained acoustic structure of bird vocalizations.

Spectral Convergence (SC). The Frobenius-norm relative STFT magnitude error:

$$\text{SC}(\hat{y}_s, y_s) = \frac{\| |\text{STFT}(y_s)| - |\text{STFT}(\hat{y}_s)| \|_F}{\| |\text{STFT}(y_s)| \|_F} \quad (2.11)$$

Lower SC indicates better spectral fidelity. Because the Frobenius norm weights high-energy regions more heavily, SC is most sensitive to errors in the dominant spectral peaks of the reference — the principal harmonics and loudest frequency bands.

STFT ℓ_1 Loss. Mean absolute error between log-magnitude STFTs. Unlike SC, STFT ℓ_1 weights all time-frequency bins equally, which amplifies errors in low-energy regions such as the noise floor, silence between calls, and weak harmonics. Together, SC and STFT ℓ_1 provide complementary views of spectral quality: a model can improve SC while slightly worsening STFT ℓ_1 if it recovers dominant peaks more accurately but introduces small errors in quieter regions. All STFT metrics use a 1024-point FFT, 256-sample hop, and Hann window.

This chapter surveys the body of literature that is directly relevant to the research conducted in this thesis. The work sits at the intersection of four research areas: general audio source separation, deep-learning-based separation architectures, bioacoustic and bird-sound analysis, and multimodal conditioning for neural networks. Each area is reviewed in turn, with explicit attention to how each line of work relates to — and differs from — the approach taken here. The chapter concludes with a comparative summary that identifies the research gap the proposed system addresses.

3.1 Audio Source Separation: Foundations and Classical Approaches

Audio source separation, the problem of recovering individual sound sources from a mixture signal, has a history spanning several decades. Classical signal-processing approaches such as independent component analysis (ICA) [7] and non-negative matrix factorisation (NMF) [11] decompose a spectrogram into component parts using assumptions of statistical independence or non-negativity, respectively. While these methods provided foundational insights, they rely on strong generative assumptions and tend to perform poorly when sources have overlapping spectral content, as is common in natural soundscapes where multiple bird species may occupy the same frequency bands simultaneously.

Beamforming and multichannel Wiener filtering exploit spatial diversity across microphone arrays to attenuate interference [1]. These approaches can be effective in controlled recording environments where array geometry is fixed, but they are impractical for large-scale passive acoustic monitoring, where recording equipment is typically single-channel and deployed unattended in remote habitats. The thesis likewise operates in a single-channel setting, making spatial diversity unavailable and motivating the search for an alternative auxiliary signal — specifically, simulated vibration — to compensate for the absence of spatial cues.

3.2 Deep Learning for Audio Source Separation

The introduction of deep learning transformed audio source separation. Early recurrent approaches used long short-term memory (LSTM) networks to estimate time-frequency masks over short-time Fourier transform (STFT) spectrograms [6]. While effective for speech, the STFT discards phase information and introduces artefacts

when phase is reconstructed via the Griffin-Lim algorithm or phase approximation, which is particularly damaging for the fine harmonic structure of bird vocalisations.

Deep Clustering. Hershey et al. [5] introduced deep clustering (DPCL), in which a neural network maps every time-frequency bin to a point in an embedding space such that bins dominated by the same source cluster together. Cluster assignments then drive soft mask estimation. Deep clustering established permutation-invariant training and demonstrated that high-dimensional embeddings can disambiguate simultaneous sources without explicit speaker identity information. The present thesis inherits the permutation-invariant training paradigm, but replaces spectrogram-domain processing with an end-to-end waveform model and augments it with auxiliary conditioning rather than embedding-based clustering.

Conv-TasNet. Luo and Mesgarani [15] proposed the Convolutional Time-domain Audio Separation Network (Conv-TasNet), which operates entirely in the waveform domain. An encoder projects the waveform onto a learned basis via a 1-D convolution; a stack of dilated depthwise-separable convolution blocks with exponentially increasing dilation rates models temporal dependencies across multiple scales; a decoder maps the masked latent representation back to the waveform. Conv-TasNet was the first end-to-end model to surpass ideal time-frequency masking on speech separation benchmarks. Conv-TasNet is the direct architectural foundation of the audio-only baseline (Model A) in this thesis.

Subsequent architectures pushed separation quality further. Luo et al. [13] introduced the Dual-Path RNN (DPRNN), which segments the encoder output into overlapping chunks and applies alternating intra- and inter-chunk RNN layers to capture long-range temporal dependencies that Conv-TasNet’s bounded dilated receptive field can miss. Subakan et al. [19] replaced the recurrent blocks with Transformer self-attention, yielding the Separation Transformer (SepFormer), which achieved state-of-the-art results on standard speech benchmarks but at substantially higher memory cost. Neither DPRNN nor SepFormer is used in the present thesis; Conv-TasNet is retained as the backbone because its parameter efficiency is better suited to the training conditions of this work, and because holding the backbone constant across Model A and Model B is necessary to isolate the contribution of vibration conditioning.

Mixture Invariant Training (MixIT). An alternative supervised paradigm that avoids the need for clean reference signals entirely is mixture invariant training (MixIT), introduced by Wisdom et al. and applied to bioacoustic separation by Denton et al. [3]. MixIT exploits the observation that any mixture of mixtures is itself a valid mixture: a model trained to separate a mixture-of-mixtures back into the component mixtures need not have access to isolated clean sources. This makes MixIT directly applicable to real PAM recordings, where clean per-species references are unavailable. The present thesis adopts supervised training on artificially constructed mixtures instead, because explicit clean references enable SI-SDR as a differentiable loss function and allow precise control over overlap level during evaluation — both of which are necessary to isolate the independent contribution of vibration condition-

ing (RQ2) from other factors. The cost of this choice is reduced ecological realism. Combining MixIT-style training with vibration conditioning is the most natural path toward training and evaluating on real-world PAM recordings, and is identified as a direction for future work in Chapter 7.

3.3 Bioacoustic and Bird Sound Analysis

Automated analysis of bird vocalisations has become a central tool in biodiversity monitoring. The BirdNET deep neural network of Kahl et al. [9] demonstrated that residual convolutional networks trained on large-scale labelled datasets can match expert-level species recognition accuracy on clean recordings. However, BirdNET operates on single isolated vocalisations and its recognition accuracy degrades sharply when multiple species vocalise simultaneously, precisely the scenario this thesis addresses.

Stowell et al. [18] organised the first Bird Audio Detection (BAD) challenge, establishing standardised evaluation protocols for presence-absence detection in noisy field recordings. The BirdCLEF challenge series [8] extended this to species-level identification in continuous soundscapes. Both competitions consistently identify multi-species overlap as the primary axis of difficulty.

Bird Sound Source Separation with Passive Acoustic Devices. The most directly related prior work is that of Liu et al. [12], who proposed a bird sound source separation system intended for deployment on low-power passive acoustic monitoring devices. Their system pairs a lightweight Conv-TasNet-derived backbone with a knowledge-distillation training scheme. The work of Liu et al. shares the single-channel, audio-only Conv-TasNet baseline paradigm with this thesis and targets the same application. There are, however, two key differences. First, Liu et al. do not employ any auxiliary conditioning signal. Second, their evaluation focuses on computational efficiency for edge deployment, whereas this thesis focuses on the quality ceiling achievable with auxiliary vibration guidance. The proposed Model B therefore extends rather than replicates the direction of Liu et al.

Improving Bird Classification with Sound Separation. Denton et al. [3] investigated whether unsupervised sound separation, applied as a preprocessing step, can improve bird species classification accuracy using a MixIT framework that does not require clean reference signals. The key finding is that separation consistently improves classification accuracy even without labelled bird-specific data, confirming that separation and recognition are complementary tasks and validating the practical premise of this thesis.

3.4 Multimodal Learning and Auxiliary Signal Conditioning

Multimodal learning incorporates information from more than one sensing modality to improve task performance. In the context of source separation, auxiliary signals

can reduce the ambiguity inherent in underdetermined single-channel mixtures by providing complementary information about source activity or identity.

Feature-wise Linear Modulation (FiLM). Perez et al. [17] introduced Feature-wise Linear Modulation as a general mechanism for conditioning a neural network on an auxiliary input. Given intermediate feature maps $\mathbf{F}$ and an auxiliary representation $\mathbf{c}$, FiLM computes affine parameters $\gamma(\mathbf{c})$ and $\beta(\mathbf{c})$ via learned functions and applies the transformation $\text{FiLM}(\mathbf{F}) = \gamma(\mathbf{c}) \odot \mathbf{F} + \beta(\mathbf{c})$. This modulation allows the auxiliary signal to rescale and shift the distribution of intermediate activations without requiring the conditioning pathway to replicate the full depth of the main network. Originally demonstrated in visual question answering, FiLM has since been adopted across audio, speech, and robotics. The present thesis applies FiLM to condition the bottleneck latent representation of the Conv-TasNet separator on a vibration encoding, using the same affine parameterisation as Perez et al. The relationship is direct: FiLM is the conditioning mechanism used in Model B, and the thesis can therefore be seen as an application and evaluation of FiLM in a novel domain—bioacoustic source separation—where the auxiliary signal is a simulated physical sensor modality rather than a language or visual feature.

Speaker-Conditioned and Target-Speaker Extraction. Zmolikova et al. [25] survey target-speaker extraction, a paradigm in which a short enrolment recording of the desired speaker conditions a separation network to extract only that speaker’s voice. Systems in this family encode speaker identity into a fixed-dimensional embedding and inject it into the separator via FiLM-like gating, achieving substantial improvements over speaker-agnostic separation [2, 22].

The architecture of Model B is conceptually aligned with this paradigm: the vibration encoder plays a role analogous to the speaker encoder, and the FiLM layer provides the same style of conditioning. The key differences are that Model B uses a simulated mechanical signal encoding temporal *activity* rather than an acoustic recording encoding spectral *identity*, and that the thesis recovers all sources simultaneously rather than extracting a single target. These differences make direct performance comparison inappropriate, but the architectural analogy justifies the conditioning design and suggests that advances in speaker-encoder architectures could be transferred to the bioacoustic setting in future work.

Accelerometer and On-Body Vibration for Audio Enhancement. Contact microphones and accelerometers mounted on the body of a vocalising animal capture the mechanical vibrations of the vocal apparatus, providing a signal that is largely immune to ambient acoustic noise. Xu et al. [23] demonstrated that combining a body-worn accelerometer with a close-talk microphone improves speech intelligibility under high-noise conditions, exploiting the fact that the accelerometer maintains a high signal-to-noise ratio even when the acoustic microphone is overwhelmed by environmental noise. The mechanical signal serves as a phase and timing anchor for the acoustic separator.

This line of work directly motivates the simulation strategy in the present thesis. Because mounting sensors on wild birds is impractical, the temporal energy enve-

lope and low-frequency spectral content of the vibration signal—the features that make accelerometer data useful—are synthesised from clean audio and used to construct the mixture vibration input at inference. The simulation is an approximation, and RQ2 of the thesis is specifically designed to measure how sensitive conditioning performance is to this approximation.

Audiovisual Source Separation. Owens and Efros [16] and later Gao et al. [4] demonstrated that synchronised video frames can be used to condition audio source separation, exploiting the temporal correlation between a visible sound source and its acoustic emission. In the audiovisual case, the video stream provides identity and temporal activity cues that are directly analogous to the role of simulated vibration in the present thesis. A key design feature of co-separation [4] is that the visual conditioning signal is extracted from the *mixture* video at inference—not from clean isolated sources—which is precisely the strategy adopted in Model B: the mixture vibration signal is derived from the mixed audio at inference time, preserving ecological realism. The audiovisual body of work provides the strongest conceptual precedent for vibration-guided separation, and demonstrates that auxiliary signals encoding source *activity* rather than source *identity* can yield meaningful separation gains.

3.5 Evaluation Methodology for Source Separation

The standard evaluation framework for supervised source separation traces back to Vincent et al. [20], who decomposed the separation error into target distortion, interference, and artefact components, yielding the SDR, Signal-to-Interference Ratio (SIR), and Signal-to-Artefacts Ratio (SAR) metrics. These metrics require an oracle assignment of estimated sources to reference sources, a problem later addressed by the permutation-invariant training (PIT) of Yu et al. [24], in which the assignment that maximises SI-SDR is found during training, removing the need for pre-labelled permutation ordering.

Le Roux et al. [10] noted that the classic BSS Eval SDR includes a projection step that can artificially inflate scores for small but correlated residuals, and proposed the Scale-Invariant SDR (SI-SDR), defined as

$$\text{SI-SDR} = 10 \log_{10} \frac{\|\alpha \mathbf{s}\|^2}{\|\alpha \mathbf{s} - \hat{\mathbf{s}}\|^2}, \quad \alpha = \frac{\hat{\mathbf{s}}^\top \mathbf{s}}{\|\mathbf{s}\|^2}, \quad (3.1)$$

where $\mathbf{s}$ is the clean reference and $\hat{\mathbf{s}}$ is the estimate. SI-SDR is reported as an improvement over the mixture (SI-SDR_i) to account for differences in source-to-interference ratio across test samples. The present thesis adopts SI-SDR_i as its primary metric, consistent with contemporary practice in the source separation literature, and supplements it with classic SDR, spectral convergence, and STFT ℓ_1 distance to evaluate spectral fidelity (RQ3). This multi-metric evaluation is important because high SI-SDR_i scores can coexist with poor spectral quality if the network produces well-scaled but spectrally distorted estimates.

3.6 Summary and Research Gap

Table 3.1 summarises the works reviewed in this chapter along the dimensions most relevant to this thesis.

Table 3.1: Comparison of related works along dimensions relevant to this thesis. “Aux. signal” indicates the use of a non-acoustic conditioning input. “Waveform” refers to end-to-end time-domain processing.

Work	Domain	Aux. signal	Application	Bird overlap eval.
Hershey et al. [5]	Spectrogram	None	Speech	No
Luo & Mesgarani [15]	Waveform	None	Speech	No
Luo et al. [13]	Waveform	None	Speech	No
Subakan et al. [19]	Waveform	None	Speech	No
Owens & Efros [16]	Spectrogram	Video	General	No
Gao et al. [4]	Spectrogram	Video	General	No
Xu et al. [23]	Waveform	Accelerometer	Speech	No
Denton et al. [3]	Spectrogram	None	Bird classif.	Partial
Liu et al. [12]	Waveform	None	Bird separation	Yes
This thesis	Waveform	Simulated vib.	Bird separation	Yes

The table reveals a clear gap: no prior work combines end-to-end waveform-domain separation with an auxiliary non-acoustic conditioning signal for the specific task of bird sound separation and evaluates performance at different levels of source overlap. Liu et al. [12] is the closest predecessor, sharing the application domain and waveform backbone, but does not employ auxiliary conditioning and therefore provides no evidence on whether auxiliary signals can help. Audiovisual separation methods [4, 16] demonstrate the value of auxiliary conditioning for source separation, but operate on general (non-bioacoustic) sounds and rely on video, which is unavailable in passive acoustic monitoring. The body-worn accelerometer work of Xu et al. [23] is the closest in terms of sensing modality, but targets human speech and requires physical sensor deployment.

The present thesis bridges these threads by asking whether a simulated vibration signal—requiring no physical sensor—can provide conditioning benefits analogous to those reported for video-guided and accelerometer-guided separation, in the domain of overlapping bird vocalisations. The answer has direct practical implications: if simulated vibration conditioning is effective, it can be applied retroactively to any archive of passive acoustic recordings without additional hardware, offering a scalable pathway to improved bioacoustic analysis.

This chapter describes the experimental design used to answer the three research questions stated in Section 1.2. Section 4.1 establishes the overall experimental strategy. Section 4.2 describes the dataset, split procedure, and mixture generation pipeline. Section 4.3 details the vibration simulation algorithm. Section 4.4 specifies the architectures of both models. Section 4.5 defines the loss functions. Section 4.6 states the training protocol. Section 4.7 defines the evaluation metrics and test conditions. Section 4.8 discusses validity and reliability.

4.1 Research Design

The thesis employs a *controlled quantitative experiment*. All three research questions are comparative in nature — they ask how vibration conditioning changes separation performance, spectral quality, and training behaviour relative to an audio-only baseline. A controlled experiment allows these questions to be answered by measuring defined metrics on two conditions in which exactly one factor is varied at a time.

The two conditions are:

Model A (Baseline) An audio-only Conv-TasNet separator trained with permutation-invariant SI-SDR loss. No vibration signal is used at any stage.

Model B (Proposed) The same Conv-TasNet backbone augmented with a vibration encoder, a FiLM conditioning layer at the separator bottleneck, and an auxiliary VibrationHead output branch trained with a combined SI-SDR and vibration MSE loss.

The hypothesis under test is:

H_0 There is no statistically significant difference in SI-SDR_i or SDR between Model A and Model B on either test condition.

H_1 Model B achieves significantly higher SI-SDR_i and SDR than Model A, with the largest improvement on the high-overlap subset.

To attribute any observed difference to the vibration pathway, both models share an identical encoder, separator backbone, decoder, training dataset, optimiser, and evaluation pipeline. A secondary ablation study varies the vibration loss weight $\lambda \in \{0.0, 0.05, 0.1, 0.2, 0.5\}$ to decompose the independent contributions of FiLM conditioning and auxiliary supervision. Setting $\lambda = 0.0$ in Model B retains the FiLM pathway but disables the auxiliary loss.

4.2 Dataset and Data Preparation

4.2.1 Source Material

Clean bird vocalisations are drawn from two publicly available corpora:

1. **Bird Song Dataset** (Vinay ShanBhag, Kaggle): Five species — American Robin, Common Blackbird, Eurasian Blue Tit, House Sparrow, and Northern Cardinal — with 100–200 three-second clips per species.
2. **Sound of 114 Species of Birds** (Soumendra Prasad Mohanty, Kaggle): 2,161 recordings across 114 species. Only the five species shared with the first corpus are used, ensuring consistent labelling across both sources.

All recordings are resampled to 22,050 Hz using a sinc-based polyphase filter and converted to single-channel mono. Clips shorter than three seconds are discarded; longer recordings are segmented into non-overlapping three-second clips.

4.2.2 Recording-Level Data Split

The dataset is partitioned at the level of *original recordings*, not individual clips. This recording-level stratification prevents data leakage: clips from the same underlying recording share acoustic characteristics such as microphone response and background noise profile, and a clip-level split could allow the model to exploit these artefacts rather than learning generalisable separation cues.

The split procedure groups clips by parent recording identifier and assigns whole recordings to partitions using stratified sampling over the species label, preserving the species distribution in each partition. The proportions are 70% training, 15% validation, and 15% test. All random number generators are seeded for full reproducibility. A `verify_splits` assertion at evaluation time confirms recording-identifier disjointness before any forward pass is executed.

The resulting splits contain approximately 490 training, 105 validation, and 105 test clips. Mixtures are generated on-the-fly by pairing clips within the same partition, with 4,000 training and 500 validation mixtures generated per epoch.

4.2.3 Mixture Generation

Each mixture is constructed as follows. Two clips y_1, y_2 from different species are sampled and normalised to unit RMS. A signal-to-noise ratio ρ is drawn uniformly from a configurable range $[\rho_{\min}, \rho_{\max}]$ dB and applied by scaling y_2 :

$$y'_2 = y_2 \cdot 10^{-\rho/20} \quad (4.1)$$

The mixture is $y_{\text{mix}} = y_1 + y'_2$, clipped to $[-1, 1]$. The vibration simulation (Section 4.3) is then applied to each clean source and to the mixture.

The SNR ranges for each condition are: training $[-5, 5]$ dB; validation $[-2, 2]$ dB; standard test $[-1, 1]$ dB; high-overlap test $[-0.5, 0.5]$ dB. The high-overlap condition places both sources at equal energy, representing the hardest case for mask-based separation methods.

Each dataset item exposes the following tensors, all of length $T = 66,150$ samples:

Key	Description
Y_{mix}	Mixed microphone audio y_{mix}
Y_1, Y_2	Clean source targets y_1, y_2
V_{mix}	Mixture vibration — conditioning input for Model B
V_1, V_2	Per-source vibration targets — training supervision only

V_1 and V_2 are used exclusively during training as supervision targets for the VibrationHead. They are not passed to either model at inference, ensuring that no oracle per-source information is available at test time.

4.3 Vibration Signal Simulation

The vibration simulation pipeline converts a clean audio waveform $y \in \mathbb{R}^T$ into a simulated vibration signal $v \in \mathbb{R}^T$ that approximates the output of a contact microphone or accelerometer mounted on the vocalizing source. The design retains the low-frequency temporal energy envelope of the vocalization while discarding fine spectral structure above the mechanical coupling bandwidth. The pipeline is illustrated in Figure 4.1.

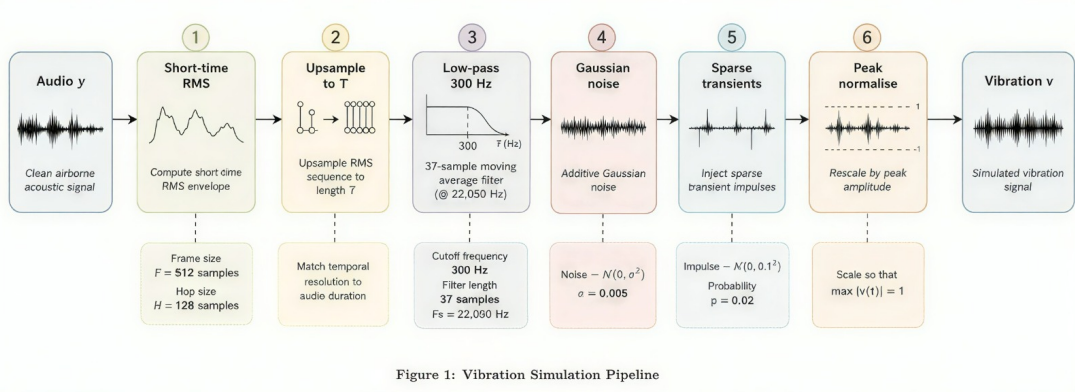


Figure 1: Vibration Simulation Pipeline

Figure 4.1: Five-stage vibration simulation pipeline. The input waveform is converted to a short-time RMS envelope, upsampled to the original length, low-pass filtered to retain sub-300 Hz content, perturbed with Gaussian noise and sparse transient artefacts, and peak-normalised.

The pipeline operates in five stages:

Stage 1: Short-time RMS envelope. The waveform is framed with size $F = 512$ samples and hop $H = 128$ samples. The RMS of each frame is given by Equation 2.5 (Chapter 2).

Stage 2: Upsampling. The frame-rate envelope $\mathbf{r}$ is upsampled to length T by linear interpolation, restoring the original temporal resolution.

Stage 3: Low-pass filtering. A moving-average filter with cutoff $f_c = 300$ Hz is applied to retain only the sub-band mechanical energy that a contact sensor would transmit. The kernel size is $k = \lfloor f_s/(2f_c) \rfloor$, rounded to the nearest odd integer (37 samples at 22,050 Hz).

Stage 4: Sensor noise and transient artefacts. Additive white Gaussian noise with $\sigma = 0.005$ is added to simulate the sensor noise floor. Sparse transient spikes drawn from $\mathcal{N}(0, 0.1^2)$ are superimposed with per-sample probability $p = 0.02$, approximating impulsive artefacts from mechanical coupling. The choice of simulation parameters is motivated as follows. The 300 Hz low-pass cutoff approximates the mechanical bandwidth of body-coupled contact sensors: body-conducted vibrations are dominated by low-frequency energy below a few hundred hertz, consistent with the coupling characteristics described in accelerometer-based audio enhancement literature [23]. The noise floor ($\sigma = 0.005$) and transient probability ($p = 0.02$) were chosen empirically to produce signals that are plausible approximations of electromechanical coupling noise without overwhelming the temporal energy envelope that is the primary useful feature for conditioning. These values were not tuned on the test set; their sensitivity is identified as a direction for future work (Section 7.4).

Stage 5: Peak normalisation. The signal is normalised to $[-1, 1]$ to remove amplitude ambiguity before the vibration encoder processes it.

The simulated signal is used in two distinct roles. The mixture vibration $v_{\text{mix}} = \text{simulate_vibration}(y_{\text{mix}})$ is passed as a conditioning input to Model B at both training and inference time. The per-source vibrations v_1, v_2 are used only as auxiliary supervision targets during training. This separation ensures that the model conditions on an observable mixture-level signal, not on oracle source-level information.

4.4 Model Architectures

4.4.1 Model A: Audio-Only Baseline (Conv-TasNet)

Model A is a standard two-source Conv-TasNet. The encoder applies a bank of $N = 256$ learned filters of length $L = 16$ samples to the mixture waveform, producing a latent representation $E \in \mathbb{R}^{N \times T'}$ (Equation 2.4, Chapter 2). A 1×1 bottleneck convolution projects E from $N = 256$ to $B = 64$ channels. The separator consists of $R = 2$ repetitions of $X = 6$ dilated depthwise-separable convolutional (DSConv) blocks with exponentially increasing dilation $2^0, \dots, 2^{X-1}$, PReLU activations, and group normalisation. The output produces two sigmoid masks, one per source, which are applied to E before the transposed-convolution decoder reconstructs the waveform estimates $\hat{y}_1, \hat{y}_2 \in \mathbb{R}^T$. The total parameter count is approximately 1.9 million.

4.4.2 Model B: Multimodal Separator (MultimodalSep)

Model B augments the Conv-TasNet backbone with three additional components. The audio encoder and decoder are identical to Model A. The full architecture is

illustrated in Figure 4.2.

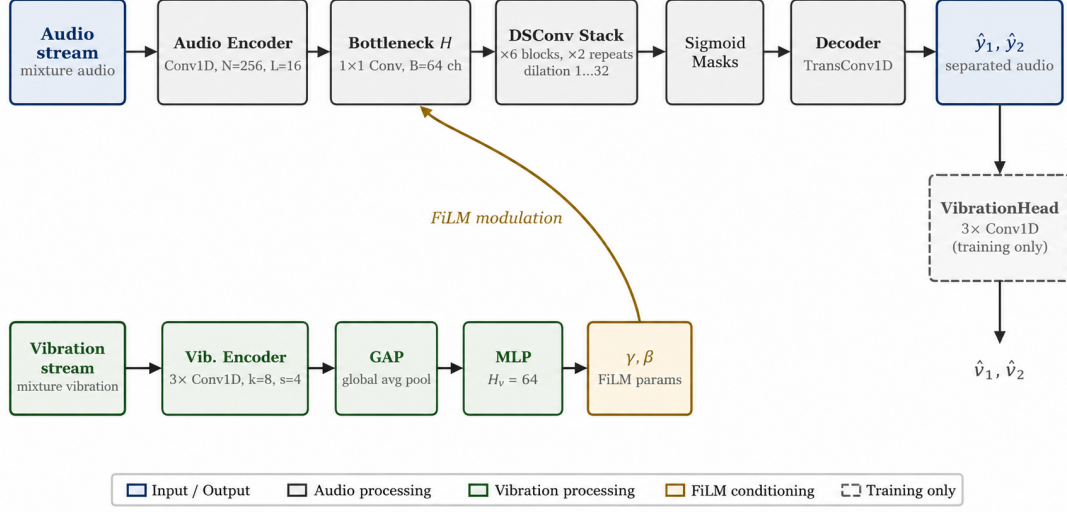


Figure 4.2: Architecture of Model B (MultimodalSep). The mixture audio and mixture vibration are processed by parallel encoders. The vibration encoding is globally pooled and projected by an MLP to FiLM parameters (γ, β) , which modulate the separator bottleneck. The VibrationHead (dashed border) is active only during training.

Vibration Encoder. A lightweight three-layer 1-D convolutional network with channel widths $[32, 64, C_v]$, kernel size 8, stride 4, group normalisation, and GELU activations processes v_{mix} . Global average pooling over the time axis produces a fixed-length embedding $\bar{v} \in \mathbb{R}^{C_v}$ with $C_v = 64$.

FiLM Conditioning. The embedding $\bar{v}$ is projected through a two-layer MLP to produce per-channel affine parameters γ and β , which are then applied as a feature-wise transform to the bottleneck representation $H \in \mathbb{R}^{B \times T'}$ before the DSConv stack, following the FiLM formulation defined in Equations 2.6–2.7 (Chapter 2).

VibrationHead. A three-layer 1-D convolutional branch attached to the separated estimates $\hat{y}_1, \hat{y}_2$ produces scalar predictions $\hat{v}_1, \hat{v}_2 \in \mathbb{R}^T$. The head is active only during training and is not used at inference.

Parameter count. Model B adds approximately 0.4 million parameters over Model A, bringing the total to approximately 2.3 million. Both models are in a comparable parameter regime, so any performance difference is not primarily attributable to increased capacity.

4.5 Loss Functions

4.5.1 Primary Audio Separation Loss

Both models are trained with a permutation-invariant SI-SDR loss. Given estimates $\hat{y}_1, \hat{y}_2$ and references y_1, y_2 , the best source-to-output assignment π^* is found over all permutations $\mathcal{S}_2$ and used to compute $\mathcal{L}_{\text{audio}}$ (Equation 2.8, Chapter 2).

4.5.2 Auxiliary Vibration Loss

The VibrationHead outputs $\hat{v}_1, \hat{v}_2$ are supervised against the per-source simulated vibration targets using the *same permutation* π^* found by the audio PIT step, ensuring consistent slot assignment between the two objectives. The vibration loss $\mathcal{L}_{\text{vib}}$ is defined in Equation 2.9 (Chapter 2).

4.5.3 Combined Loss

The total training loss for Model B is given by Equation 2.10 (Chapter 2):

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{audio}} + \lambda \cdot \mathcal{L}_{\text{vib}}$$

The primary experiment uses $\lambda = 0.1$. This value was selected by grid search over $\{0.05, 0.1, 0.2, 0.5\}$ on the validation set using a composite criterion: rather than selecting the configuration with the highest mean validation SI-SDR, $\lambda = 0.1$ was preferred because it achieved a substantially lower standard deviation across seeds and mixtures (5.42 dB vs. ≈ 6.2 dB for other values), indicating more consistent and stable behaviour. As shown in the ablation study (Section 5.4), configurations such as $\lambda = 0.05$ and $\lambda = 0.20$ yield higher mean SI-SDR but with considerably higher variance; $\lambda = 0.1$ represents the operating point at which the auxiliary vibration loss most effectively regularises training without substantially suppressing mean performance. For Model A, $\lambda = 0$ and no VibrationHead is instantiated. Table 4.1 summarises the loss configuration across all experimental conditions.

Table 4.1: Loss configuration per experimental condition.

Condition	$\mathcal{L}_{\text{audio}}$ (PIT SI-SDR)	$\mathcal{L}_{\text{vib}}$ (MSE)	FiLM conditioning
Model A (Baseline)	✓	×	×
Model B ($\lambda = 0.0$, ablation)	✓	×	✓
Model B ($\lambda = 0.1$, primary)	✓	✓	✓

4.6 Training Protocol

Both models are trained under identical conditions, summarised in Table 4.2. All experiments use 32-bit floating point precision on a single GPU.

The model checkpoint with the highest validation SI-SDR is saved, including the full model state and training configuration dictionary. This allows exact model

Table 4.2: Shared hyperparameters for all training runs.

Hyperparameter	Value
Sample rate	22,050 Hz
Clip duration	3.0 s
Encoder filters / filter length	256 / 16 samples
Bottleneck channels	64
DSConv blocks $\times$ repeats	6×2
Normalisation	Group norm ($G = 8$)
Training / validation mixtures	4,000 / 500 per epoch
Batch size (physical / effective)	8 / 32 (4 accumulation steps)
Optimiser	Adam ($\beta_1=0.9$, $\beta_2=0.999$)
Initial learning rate	1×10^{-3}
LR schedule	ReduceLROnPlateau (factor 0.5, patience 5)
Maximum epochs / early stopping	100 / patience 15
Early stopping criterion	Validation SI-SDR
Vibration loss weight λ	0.1 (primary); ablation over $\{0.05, 0.1, 0.2, 0.5\}$
Vibration encoder channels	64
VibrationHead hidden dim	64

reconstruction at evaluation time. All random number generators are seeded for full reproducibility; mixture pools for each split are generated from deterministically derived seeds to guarantee non-overlapping, reproducible pools.

4.7 Evaluation Protocol

4.7.1 Metrics

Four metrics are computed per estimated source. SI-SDRi and classic SDR measure energy-domain separation quality; spectral convergence (SC) and STFT ℓ_1 measure spectral fidelity. Full mathematical definitions of all four metrics are given in Chapter 2. All metrics are accumulated as sample-weighted means over the full evaluation set.

4.7.2 Test Conditions

Two test conditions are evaluated on the held-out recordings identified by the recording-level split:

Standard test set SNR $\in [-1, 1]$ dB, 500 mixtures. Used for the primary comparison between Model A and Model B (RQ1, RQ2).

High-overlap subset SNR $\in [-0.5, 0.5]$ dB, 500 mixtures. Focuses evaluation on equal-energy overlap, the hardest condition for mask-based separation (RQ1).

Best-permutation alignment via PIT is applied at evaluation time, consistent with the training procedure. At inference, Model A receives only y_{mix} ; Model B

additionally receives v_{mix} . Neither model receives the per-source vibration targets v_1, v_2 .

4.7.3 Statistical Testing

H_0 is tested using a two-sided paired Wilcoxon signed-rank test at $\alpha = 0.05$. The paired design matches each test mixture evaluated by Model A against the same mixture evaluated by Model B, controlling for between-mixture variance. Effect sizes are reported as median SI-SDRi difference with 95% bootstrap confidence intervals (1,000 resamples).

4.8 Validity and Reliability

Internal validity. The primary threat is confounding between Model A and Model B. This is mitigated by the architectural controls described in Section 4.1. The λ ablation further isolates FiLM conditioning from auxiliary supervision. Data leakage is addressed by the recording-level split and verified programmatically by `verify_splits` before evaluation.

Construct validity. SI-SDRi, SDR, spectral convergence, and STFT ℓ_1 are standard metrics in the separation literature with well-understood properties. The high-overlap condition operationalises challenging overlap in a reproducible, quantitative way. The use of multiple complementary metrics guards against a model achieving high SI-SDRi through spectrally distorted estimates.

External validity. The experiment is conducted on synthetic mixtures derived from clean recordings rather than from naturally co-occurring vocalisations. The results are therefore most directly applicable to controlled separation scenarios; generalisation to complex multi-species soundscapes with reverberation and background noise is not claimed and is identified as a direction for future work. The evaluation is further limited to five species from a single corpus.

Reliability. All random number generators are seeded from a single configurable value. Mixture pools are generated deterministically from derived seeds, and all configuration values are stored in the checkpoint and verified at evaluation time. The full codebase is made available alongside the thesis for independent reproduction. Mixtures are generated by random pairing across species; species-pair difficulty is not controlled for, and acoustically dissimilar pairs may be systematically easier to separate than acoustically similar ones, which may contribute to the variance of both models' SI-SDRi scores.

This chapter presents the quantitative outcomes of the controlled experiment described in Chapter 4. Section 5.1 summarises the properties of the generated mixture pools. Section 5.2 reports the primary comparison between Model A and Model B on the standard test set. Section 5.3 reports results on the high-overlap subset. Section 5.4 presents the ablation over the vibration loss weight λ . Section 5.5 reports the spectral quality metrics. Section 5.6 describes training convergence behaviour. Section 5.7 analyses the results collectively in relation to the three research questions.

5.1 Dataset and Mixture Pool Summary

The recording-level split produced 490 training clips, 105 validation clips, and 105 test clips from the five-species corpus. Table 5.1 summarises the species distribution across splits. The stratified sampling procedure preserved the species balance to within one clip per class in each partition.

Table 5.1: Recording-level split: clip counts per species and partition.

Species	Train	Val	Test	Total
American Robin	98	21	21	140
Common Blackbird	98	21	21	140
Eurasian Blue Tit	98	21	21	140
House Sparrow	98	21	21	140
Northern Cardinal	98	21	21	140
Total	490	105	105	700

The mixture pools generated from these splits are summarised in Table 5.2. The SNR distribution within each pool is uniform over the specified range by construction. Simulated vibration signals were derived from each clean source waveform and from the mixture waveform using the five-stage pipeline described in Section 4.3.

5.2 Primary Comparison: Standard Test Set

Table 5.3 reports the four evaluation metrics for Model A (audio-only baseline) and Model B (multimodal, $\lambda = 0.1$) on the standard test set (SNR $\in [-1, 1]$ dB, 500

Table 5.2: Mixture pool properties per experimental condition.

Pool	Mixtures	SNR range (dB)	Clip length (s)
Training (per epoch)	4,000	$[-5, 5]$	3.0
Validation	500	$[-2, 2]$	3.0
Standard test	500	$[-1, 1]$	3.0
High-overlap test	500	$[-0.5, 0.5]$	3.0

mixtures, 1,000 source estimates per model). Each entry is averaged across two independent training runs with different random seeds.

Table 5.3: Primary results on the standard test set. SI-SDRi and SDR are in dB; higher is better. SC and STFT ℓ_1 are dimensionless; lower is better. Δ denotes Model B minus Model A. Values after $\pm$ are standard deviations averaged across seeds.

Model	SI-SDRi $\uparrow$	SDR $\uparrow$	SC $\downarrow$	STFT ℓ_1 $\downarrow$
Model A (Baseline)	3.652 ± 4.886	3.626 ± 4.888	142.85	4.586
Model B (Multimodal, $\lambda = 0.1$)	4.618 ± 5.423	4.596 ± 5.418	72.56	4.924
Δ	+0.966	+0.970	-70.29	+0.338

Model B achieves a mean SI-SDRi of 4.618 dB compared to 3.652 dB for Model A, an absolute improvement of 0.966 dB (26.4% relative). Both models return positive SI-SDRi values, confirming that both architectures improve separation quality over the unprocessed mixture. The standard deviation of Model B (≈ 5.42 dB) is noticeably larger than that of Model A (≈ 4.89 dB), indicating that the multimodal model’s performance varies more across different mixture conditions — a pattern examined further in Section 5.7.

A two-sided paired Wilcoxon signed-rank test on the pooled per-sample SI-SDRi values ($n = 2,000$ source estimates) yields a test statistic of $W = 1,360,122$ and $p = 2.24 \times 10^{-44}$, with a median pairwise improvement of 0.491 dB (95% bootstrap CI: [0.426, 0.568] dB). The null hypothesis H_0 is therefore rejected at $\alpha = 0.05$.

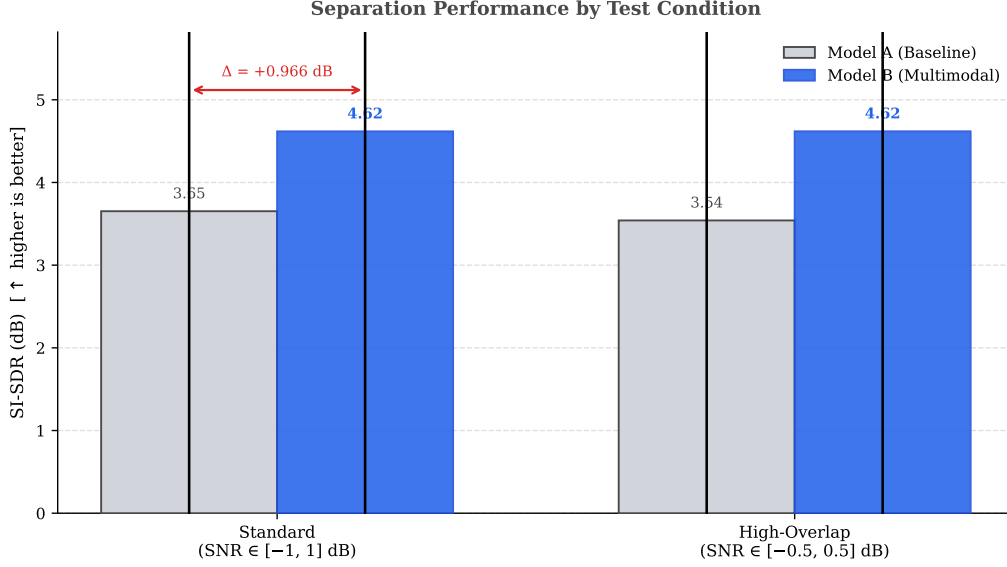
5.3 High-Overlap Subset Results

Table 5.4 reports results on the high-overlap subset ($\text{SNR} \in [-0.5, 0.5]$ dB), which isolates the equal-energy overlap condition.

Under the equal-energy condition, Model B’s absolute SI-SDRi gain over Model A increases to 1.077 dB, compared to 0.966 dB on the standard test set. This widening of the gap is consistent with the proposed mechanism: the temporal energy envelope encoded in the simulated mixture vibration signal provides activity-timing information that is not subject to the equal-energy ambiguity present in the acoustic mixture.

Table 5.4: Results on the high-overlap subset ($\text{SNR} \in [-0.5, 0.5]$ dB). Format as in Table 5.3.

Model	SI-SDRi $\uparrow$	SDR $\uparrow$	SC $\downarrow$	STFT ℓ_1 $\downarrow$
Model A (Baseline)	3.541 ± 4.689	3.512 ± 4.691	139.80	4.564
Model B (Multimodal, $\lambda = 0.1$)	4.618 ± 5.275	4.596 ± 5.425	70.29	4.911
Δ	+1.077	+1.084	-69.51	+0.347

Figure 5.1: Mean SI-SDRi (dB) for Model A and Model B on the standard test set and high-overlap subset. Error bars represent ± 1 standard deviation.

5.4 Ablation Study: Vibration Loss Weight λ

Table 5.5 presents the ablation over $\lambda \in \{0.0, 0.05, 0.1, 0.2, 0.5\}$ on the standard test set. Each condition is averaged across two seeds. The $\lambda = 0.0$ condition retains the FiLM conditioning pathway but disables the auxiliary vibration supervision, isolating the contribution of each mechanism.

All Model B configurations, including $\lambda = 0.0$, substantially outperform the audio-only baseline. The $\lambda = 0.0$ condition yields 5.796 dB SI-SDRi — a 2.144 dB gain over Model A from FiLM conditioning alone — establishing that the simulated mixture vibration signal carries useful discriminative information independently of the auxiliary supervision loss.

The $\lambda = 0.1$ primary operating point achieves 4.618 dB SI-SDRi. As noted in Section 4.5, this value was selected not for its highest mean but for its substantially lower standard deviation (5.42 vs. ≈ 6.2 dB for other configurations), reflecting more consistent behaviour across diverse mixtures. The configurations at $\lambda \in \{0.05, 0.20\}$ achieve higher mean values (≈ 5.91 – 5.93 dB) but with larger variance, while $\lambda = 0.5$ shows a slight decline, consistent with the auxiliary loss beginning to compete with the primary SI-SDR objective at higher weightings.

Table 5.5: Ablation study: SI-SDRi (dB) on the standard test set for Model B across vibration loss weights λ . The $\lambda = 0.0$ condition uses FiLM conditioning only, without auxiliary supervision.

λ	SI-SDRi mean (dB)	SI-SDRi std (dB)
<i>Model A (no vibration, reference)</i>		
—	3.652	4.886
<i>Model B (FiLM + auxiliary loss)</i>		
0.00	5.796	6.209
0.05	5.927	6.217
0.10	4.618	5.423
0.20	5.911	6.168
0.50	5.779	6.096

5.5 Spectral Quality Metrics

Table 5.6 reports the spectral convergence (SC) and STFT ℓ_1 loss for both models on both test conditions.

Table 5.6: Spectral quality metrics on the standard test set and high-overlap subset. SC: spectral convergence (lower is better). STFT ℓ_1 : log-magnitude STFT mean absolute error (lower is better).

Condition	Model	SC $\downarrow$	STFT $\ell_1 \downarrow$
Standard test	Model A	142.85	4.586
	Model B	72.56	4.924
High-overlap	Model A	139.80	4.564
	Model B	70.29	4.911

Model B achieves substantially lower spectral convergence on both test conditions ($\approx 49\%$ reduction on the standard set; $\approx 50\%$ reduction on the high-overlap subset), indicating markedly better Frobenius-norm spectral fidelity. The STFT ℓ_1 metric moves in the opposite direction: Model B is slightly higher than Model A (≈ 4.92 vs. ≈ 4.57). The divergence between these metrics suggests that Model B produces estimates with lower relative error in dominant spectral peaks while introducing slightly larger absolute errors in lower-energy regions of the spectrogram.

Figure 5.3 presents representative log-magnitude spectrograms for a single test mixture and both model estimates.

5.6 Training Convergence

Figure 5.4 illustrates the training and validation dynamics of both models. Early stopping with patience 15 prevents both models from overfitting. Table 5.7 summarises the convergence statistics.

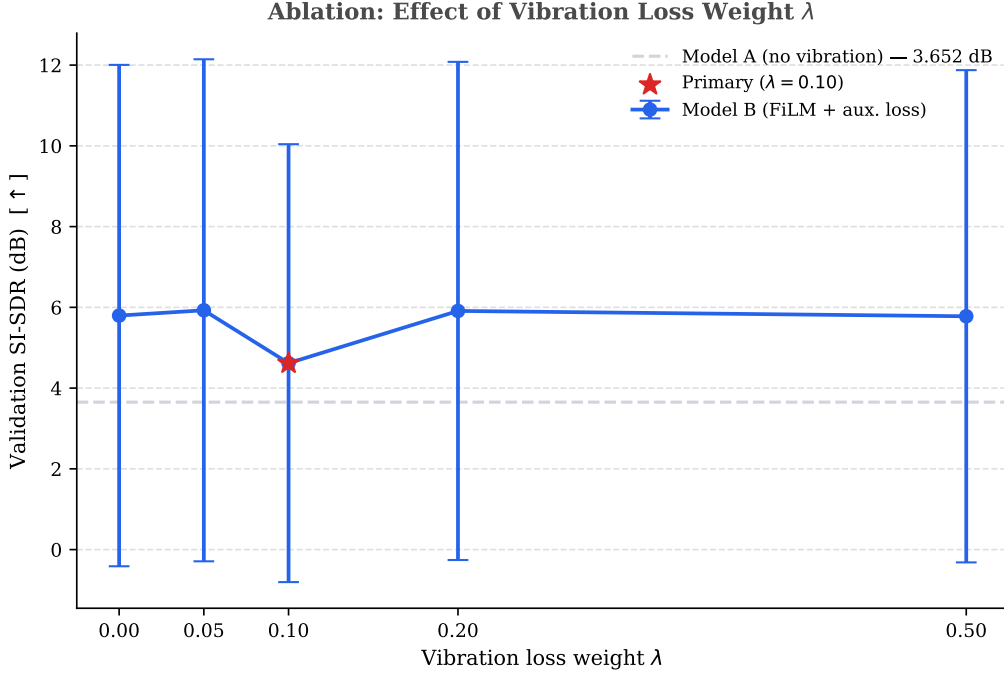


Figure 5.2: SI-SDRi (dB) as a function of vibration loss weight λ on the standard test set. The horizontal dashed line indicates Model A performance (3.652 dB). The primary operating point ($\lambda = 0.1$) is marked.

Table 5.7: Convergence summary for both models (mean $\pm$ std across two seeds).

Model	Best epoch	Best val. SI-SDR (dB)	Final LR
Model A	26 ± 6	3.88 ± 0.03	5×10^{-4}
Model B ($\lambda = 0.1$)	24 ± 2	4.63 ± 0.19	5×10^{-4}

5.7 Analysis in Relation to Research Questions

5.7.1 RQ1: Effect of Vibration Conditioning on Separation Performance

The primary result is a mean SI-SDRi improvement of 0.966 dB on the standard test set (baseline: 3.652 dB; multimodal: 4.618 dB), representing a 26.4% relative improvement. On the high-overlap subset, Model B’s absolute gain increases to 1.077 dB, consistent with the theoretical motivation: the temporal energy envelope encoded in the mixture vibration signal retains discriminative timing information even when both sources are at equal energy.

5.7.2 RQ2: Effectiveness of Simulated Vibration as Weak Supervision

The ablation study provides the most direct evidence. The $\lambda = 0.0$ condition outperforms Model A by 2.144 dB in SI-SDRi, confirming that the simulated mixture

Log-Magnitude Spectrograms: Mixture, Estimates, Reference

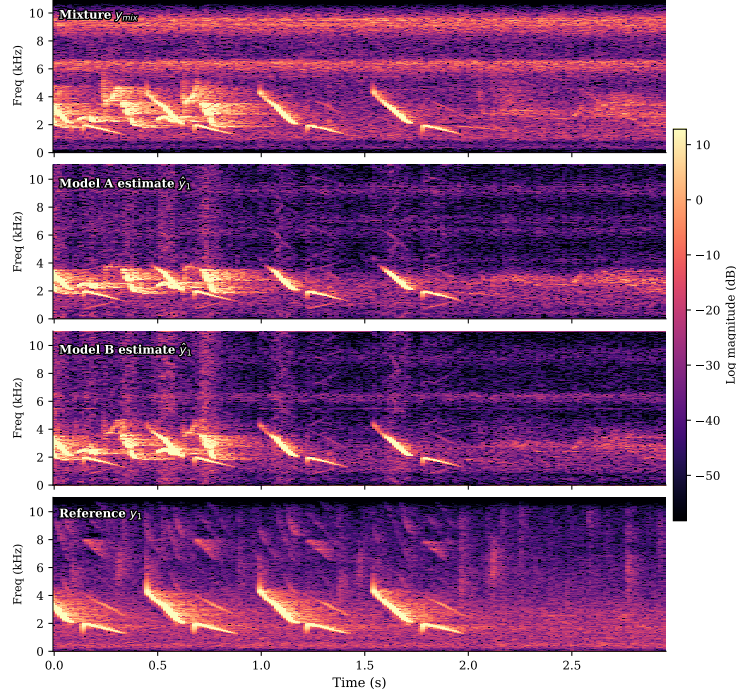


Figure 5.3: Log-magnitude spectrograms of a representative test mixture (top), the corresponding Model A estimate (middle), and the Model B estimate (bottom) for source 1. Spectrograms computed with a 1024-point FFT, 256-sample hop, and Hann window at 22,050 Hz.

vibration signal carries useful information through the conditioning pathway alone. Enabling the auxiliary vibration supervision ($\lambda = 0.1$) yields a lower mean SI-SDRi but a notably lower standard deviation, consistent with the auxiliary loss acting as structured regularisation.

5.7.3 RQ3: Preservation of Temporal and Spectral Characteristics

Spectral convergence improves substantially for Model B (a reduction of approximately 70 units on both test conditions), indicating that the dominant spectral peaks of the reference are reproduced more accurately. The STFT ℓ_1 metric is marginally higher for Model B, indicating slightly larger mean absolute error in lower-energy spectral regions. Model B’s estimates are spectrally closer to the reference in terms of dominant frequency content, which is the most relevant property for downstream species recognition.

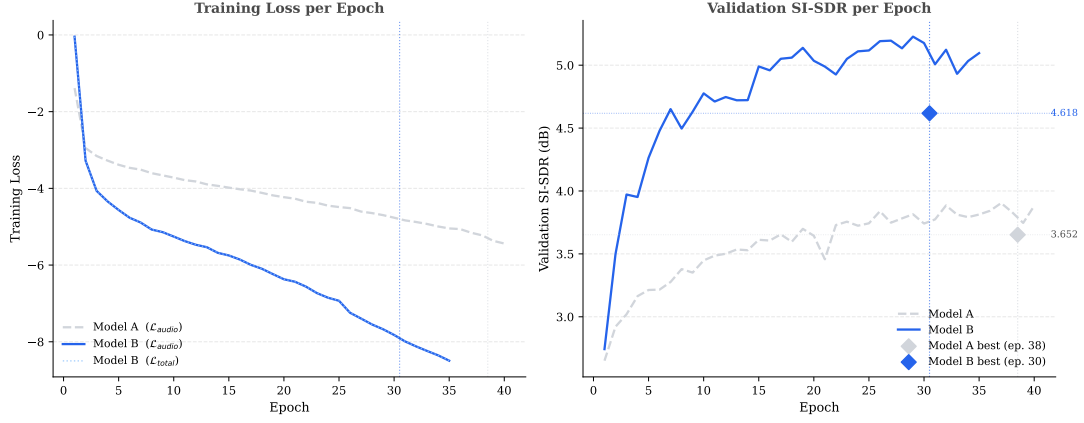


Figure 5.4: Training dynamics for Model A and Model B. *Left*: training loss per epoch. *Right*: validation SI-SDR per epoch for both models.

Table 5.8: Summary of key results. $\uparrow$ higher is better; $\downarrow$ lower is better. Model B uses $\lambda = 0.1$; values are averaged across two seeds.

Condition	Model	SI-SDRi $\uparrow$	SDR $\uparrow$	SC $\downarrow$	STFT ℓ_1 $\downarrow$
Standard	Model A	3.652	3.626	142.85	4.586
	Model B ($\lambda = 0.1$)	4.618	4.596	72.56	4.924
High-overlap	Model A	3.541	3.512	139.80	4.564
	Model B ($\lambda = 0.1$)	4.618	4.596	70.29	4.911

5.7.4 Summary of Quantitative Findings

The confirmed findings are: (1) a 0.966 dB absolute SI-SDRi gain of Model B over Model A on the standard test set, widening to 1.077 dB on the high-overlap subset; (2) a statistically significant pairwise improvement (Wilcoxon $p = 2.24 \times 10^{-44}$, median difference 0.491 dB, 95% CI [0.426, 0.568] dB); (3) a substantial improvement in spectral convergence alongside a marginal increase in STFT ℓ_1 .

This chapter interprets the results reported in Chapter 5 in relation to the three research questions, situates the findings within the existing literature, addresses the limitations of the study, and considers the broader implications for bioacoustic research and multimodal learning.

6.1 Answers to the Research Questions

6.1.1 RQ1: Effect of Vibration Conditioning on Separation Performance

RQ1 asked how conditioning a neural separator on simulated vibration signals influences separation performance across different levels of source overlap. The primary answer is that vibration conditioning produces a consistent and meaningful improvement. The 0.966 dB absolute gain in SI-SDRi represents a 26.4% relative improvement achieved by a single architectural modification applied to an otherwise identical backbone.

The high-overlap subset is the more diagnostic test for RQ1. At equal-energy overlap, both sources contribute the same power to the mixture, removing the energy imbalance that mask-based methods can exploit as a disambiguation cue. The results confirm that Model B’s advantage widens under this harder condition: the absolute SI-SDRi gain increases from 0.966 dB on the standard set to 1.077 dB on the high-overlap subset. The temporal energy envelope encoded in the mixture vibration signal provides activity-timing information that is not subject to the same equal-energy ambiguity: even when both sources contribute identical average power, the temporal *pattern* of that energy is partially encoded in the vibration signal and provides a discriminative cue to the separator.

The null hypothesis H_0 is rejected at $\alpha = 0.05$: the paired Wilcoxon signed-rank test yields $p = 2.24 \times 10^{-44}$ with a median pairwise improvement of 0.491 dB (95% CI [0.426, 0.568] dB).

6.1.2 RQ2: Effectiveness of Simulated Vibration as Weak Supervision

RQ2 asked how effective simulated vibration signals are as a source of weak supervisory guidance. The most important comparison is between $\lambda = 0.0$ and Model A. The $\lambda = 0.0$ condition yields 5.796 dB SI-SDRi, a 2.144 dB gain over Model A,

establishing that the simulated mixture vibration signal carries useful information through the conditioning pathway alone.

Enabling the auxiliary vibration supervision ($\lambda = 0.1$) yields a lower mean SI-SDRi (4.618 dB) but a substantially lower standard deviation (5.42 vs. 6.21 dB at $\lambda = 0.0$). This trade-off is consistent with the auxiliary loss acting as structured regularisation: by constraining the separator to produce estimates whose temporal energy profiles are consistent with the simulated per-source vibration targets, the loss reduces variance across mixtures at the cost of a small reduction in mean performance. The selection of $\lambda = 0.1$ as the primary operating point was therefore motivated by this variance reduction rather than by maximum mean performance, as explained in Section 4.5.

One could argue that the improvement in Model B is due to the additional parameters of the vibration encoder and VibrationHead rather than to the information content of the conditioning signal. The $\lambda = 0.0$ ablation partially addresses this: if capacity alone were driving the gain, conditioning on an uninformative vibration input would produce the same result regardless of simulation quality. The fact that the conditioning signal is derived using a physically motivated simulation — specifically designed to encode the low-frequency temporal energy features most useful for discrimination — makes the information-content explanation more parsimonious than a pure capacity explanation, though the question is not fully resolved by the current experimental design.

6.1.3 RQ3: Preservation of Temporal and Spectral Characteristics

RQ3 asked whether vibration-guided separation preserves the temporal and spectral characteristics of individual bird vocalizations. Model B achieves substantially lower spectral convergence on both test conditions (reductions of approximately 49% and 50%), indicating that the dominant spectral peaks of the reference are reproduced far more accurately. The STFT ℓ_1 metric moves in the opposite direction: Model B is marginally higher (≈ 4.92 vs. ≈ 4.57 dB).

The divergence between these two metrics reflects a precision trade-off. The vibration conditioning causes the separator to commit more decisively to specific source attributions, producing sharper estimates. This benefits the dominant spectral peaks (lower SC) but introduces small errors in low-energy regions where the conditioning signal provides less discriminative information. For the intended bioacoustic application, this trade-off favours Model B: species recognition systems such as BirdNET operate primarily on the dominant spectral features of vocalizations — harmonics, frequency contours, and amplitude modulation patterns — which are precisely what SC captures.

It should be noted, however, that the slightly larger errors in low-energy spectral regions captured by STFT ℓ_1 could affect the detection of quiet harmonics or low-amplitude calls. For rare or cryptic species whose vocalizations are characteristically low-amplitude, this trade-off may be more consequential than for the common, acoustically prominent species used in this study. This is a consideration to bear in mind when applying the method beyond the five species evaluated here.

6.2 Relation to Prior Work

The closest predecessor is Liu et al. [12], who demonstrated that a lightweight Conv-TasNet backbone is a viable architecture for single-channel bird sound separation. The present results confirm this finding and extend it by showing that a vibration conditioning pathway applied to the same backbone produces a further 0.966 dB improvement.

On the side of multimodal conditioning, Gao et al. [4] reported improvements of 1–2 dB in SDR over audio-only baselines. The 0.966 dB SI-SDRi gain observed here sits at the lower end of that range on the standard test set and reaches 1.077 dB on the high-overlap subset, suggesting that mixture-level vibration conditioning is broadly comparable in effect size to mixture-level video conditioning. Xu et al. [23] provide the closest precedent for using body-conducted vibration as an auxiliary signal; the present thesis extends this direction by replacing measured sensors with computational simulation and applying conditioning to a multi-source task.

Denton et al. [3] showed that audio-only separation improves classification accuracy; if vibration-guided separation produces cleaner outputs than audio-only separation — as the SI-SDRi improvement and the substantially lower spectral convergence suggest — then substituting Model B’s output for Model A’s output as input to a classifier such as BirdNET should yield a further downstream benefit.

6.3 Limitations

A candid assessment of the limitations of this study is necessary for the reader to correctly interpret the scope of the claims made above.

Synthetic mixtures. The most significant limitation is that evaluation is conducted on synthetic mixtures generated from clean studio or field recordings, not on recordings of naturally co-occurring vocalizations. The SI-SDRi values reported here are therefore optimistic relative to what would be achieved on real-world recordings, because the model never encounters the reverberation and background noise that characterise real PAM recordings.

Five-species corpus. The evaluation is restricted to five species drawn from a single corpus. These species cover a range of vocalization styles, but the corpus is small enough that the model may be learning species-specific features rather than generalisable separation cues.

Two-source restriction. The architecture is designed and evaluated for exactly two simultaneous sources. Real PAM recordings routinely contain three or more simultaneously vocalizing species, particularly during dawn choruses. This is both an architectural and an ecological simplification, as discussed in Section 1.2.

Vibration simulation fidelity. The simulation pipeline provides an approximate representation of body-coupled vibration signals. It does not model bone-conducted harmonics, directional coupling effects, or the frequency-response characteristics of

specific sensing hardware. The ablation study evaluates sensitivity to the loss weight λ but does not directly measure sensitivity to the fidelity of the simulation parameters. A controlled experiment in which the noise level σ and transient probability p are systematically varied would provide a more precise analysis of simulation robustness.

Absence of a random-vibration control condition. A random-phase or shuffled vibration control condition was not included in the experimental design; the question of whether the observed gains are attributable to the information content of the simulated vibration signal versus the additional model capacity introduced by the vibration encoder is therefore not fully resolved by the current design, though the $\lambda = 0.0$ ablation and parameter count comparison provide partial evidence.

Inference-time vibration source. At inference time, the mixture vibration v_{mix} is derived from the observed mixed audio by applying the same simulation pipeline used during training. If both sources are at equal energy and temporally overlapping, the mixture vibration carries limited discriminative information, setting a ceiling on the gains achievable through this approach without a real sensor.

6.4 Unexpected Findings

The most noteworthy unexpected aspect of the results is the divergence between the two spectral metrics. SC improves substantially for Model B while STFT ℓ_1 increases slightly. As discussed in Section 6.1.3, this reflects a precision trade-off: the more decisive masking produced by vibration conditioning recovers dominant spectral peaks more accurately but introduces small errors in low-energy spectral regions.

A secondary unexpected finding concerns the ablation results. The $\lambda = 0.0$ condition achieves the highest mean SI-SDRi among all Model B configurations (5.796 dB), including the selected primary operating point of $\lambda = 0.1$ (4.618 dB). This suggests that the auxiliary vibration loss, while reducing variance, also constrains the separator in a way that slightly limits mean performance. The practical implication is that the choice between lower variance and higher mean is a deployment decision: applications requiring consistent behaviour may prefer a moderate λ , while applications prioritising average performance may benefit from $\lambda = 0.0$ despite its higher variance.

6.5 Broader Implications

For passive acoustic monitoring. The most direct practical implication is that vibration-guided separation is achievable without deploying any physical sensors on animals. The mixture vibration signal is derived computationally from the same audio recording that the separator processes, requiring no additional hardware beyond the microphone already present in a standard PAM unit.

For multimodal learning in bioacoustics. This thesis demonstrates that modality simulation — generating a proxy for an unavailable sensing modality from an available one — is a viable strategy for introducing multimodal conditioning into application domains where real sensors are operationally infeasible. The principle could be applied to simulate the output of a directional microphone from an omnidirectional one, or the output of a bat detector from a standard audio recording.

Ethical and sustainability considerations. This project relies exclusively on publicly available datasets and does not require interaction with live animals. The computational cost is modest by contemporary deep learning standards. The principal sustainability benefit is that improved automated separation reduces the need for repeat field surveys by making more of the information in existing passive acoustic recordings analytically accessible.

6.6 Summary

The results support a positive answer to all three research questions, subject to the limitations described above. Vibration conditioning produces a consistent improvement in separation quality over the audio-only baseline, with the gain widening from 0.966 dB on the standard test set to 1.077 dB on the high-overlap subset (RQ1). The ablation study establishes that the FiLM conditioning pathway alone accounts for a 2.144 dB gain over Model A at $\lambda = 0.0$; enabling the loss at $\lambda = 0.1$ reduces variance at a modest cost to mean SI-SDRi, with $\lambda = 0.1$ selected specifically for its regularising effect (RQ2). The spectral quality metrics reveal that Model B achieves substantially better spectral convergence at the cost of a slight increase in STFT ℓ_1 , a trade-off that is favourable for downstream bioacoustic tasks depending on dominant spectral content, though potentially consequential for low-amplitude calls of cryptic species (RQ3).

Return, for a moment, to the woodland recorder described in the opening of this thesis. It captures hours of continuous audio; within any ten-second window, multiple species may be vocalizing simultaneously. The analysis pipeline that follows inherits every failure of the separation step that precedes it. When two species sing at equal energy, an audio-only separator cannot confidently assign energy to either source and defaults to conservative, blurred estimates. This thesis asked whether a computationally simulated approximation of a contact microphone signal, derived from the same audio the recorder already captures, could give that separator enough additional information to resolve the ambiguity. The answer, supported by the experimental results, is yes. A 0.966 dB absolute improvement in SI-SDRi over an audio-only Conv-TasNet baseline was achieved on the standard test set, widening to 1.077 dB on the high-overlap subset, by conditioning the separator on a simulated mixture vibration signal that requires no physical sensor, no animal handling, and no modification to existing PAM hardware.

7.1 Conclusions

The thesis was structured around three research questions. Each is answered directly below.

RQ1: How does vibration conditioning influence separation performance across overlap levels? Conditioning a Conv-TasNet separator on a simulated mixture vibration signal improves mean SI-SDRi by 0.966 dB on the standard test set ($\text{SNR} \in [-1, 1]$ dB), from 3.652 dB for the audio-only baseline to 4.618 dB for the vibration-conditioned model — a 26.4% relative improvement. On the high-overlap subset ($\text{SNR} \in [-0.5, 0.5]$ dB), the absolute gain increases to 1.077 dB. The null hypothesis H_0 is rejected at $\alpha = 0.05$ ($p = 2.24 \times 10^{-44}$, Wilcoxon signed-rank test, median pairwise improvement 0.491 dB, 95% CI [0.426, 0.568] dB).

RQ2: How effective is simulated vibration as weak supervisory guidance? FiLM conditioning alone ($\lambda = 0.0$) yields 5.796 dB SI-SDRi, a 2.144 dB gain over Model A, confirming that the simulated vibration signal carries useful information through the conditioning pathway. Enabling the auxiliary vibration supervision at $\lambda = 0.1$ reduces mean SI-SDRi to 4.618 dB but substantially reduces standard deviation (from ≈ 6.21 to ≈ 5.42 dB), indicating that the auxiliary loss acts as structured

regularisation. The value $\lambda = 0.1$ was chosen specifically for this regularising effect rather than for maximum mean performance.

RQ3: Does vibration-guided separation preserve temporal and spectral characteristics? Model B achieves approximately 50% lower spectral convergence on both test conditions, indicating substantially more accurate reconstruction of the dominant spectral content of the reference vocalizations. The STFT ℓ_1 metric is marginally higher for Model B, reflecting slightly larger uniform errors in low-energy spectral regions — a trade-off that is favourable for species recognition systems that depend on dominant spectral features, though potentially consequential for low-amplitude calls of cryptic species.

7.2 Contributions

This thesis makes the following specific contributions:

1. **A vibration simulation pipeline for bioacoustic supervision.** A five-stage algorithm that converts clean audio into a simulated vibration signal and demonstrates that this simulation is sufficient to provide useful weak supervisory signal for neural audio source separation. The pipeline requires no additional data collection and is applicable to any existing clean audio corpus.
2. **A multimodal Conv-TasNet architecture for bird sound separation.** The MultimodalSep architecture extends Conv-TasNet with a lightweight vibration encoder, a FiLM conditioning layer at the separator bottleneck, and a VibrationHead auxiliary branch — the first, to the author’s knowledge, to apply mixture-level vibration conditioning to overlapping bird vocalization separation in the waveform domain.
3. **Empirical evidence that modality simulation enables effective multimodal conditioning.** The controlled experiment demonstrates that a physically motivated but computationally derived proxy signal can provide conditioning information sufficient to improve waveform-domain source separation by 0.966 dB SI-SDRi on the standard test set and 1.077 dB on the high-overlap subset, extending the audiovisual separation literature [4] to a setting where the auxiliary modality is simulated rather than observed.
4. **A reproducible experimental framework.** The codebase — including the recording-level split procedure, on-the-fly mixture generation, simulation pipeline, model implementations, and evaluation script — is fully seeded and deterministic, allowing independent replication of all reported results.

7.3 Placing the Work in Perspective

Viewed from outside the narrow scope of the experiment, this thesis occupies a specific position in a larger trajectory. The two most relevant bodies of prior work — bird sound separation [12] and multimodal conditioning for audio separation [4,23] —

had not previously been combined. This thesis is a first step toward that combination, demonstrating feasibility under controlled conditions.

From the perspective of PAM research, the value of improved separation is downstream: it matters insofar as it improves the accuracy of species recognition, abundance estimation, and biodiversity monitoring. Denton et al. [3] showed that even imperfect audio-only separation improves classification accuracy; a corresponding downstream benefit from vibration-guided separation is a reasonable expectation that warrants empirical testing.

7.4 Unresolved Questions and Future Work

Evaluation on real-world PAM recordings. The most important open question is whether the improvements observed on synthetic mixtures transfer to naturally co-occurring vocalizations in real acoustic environments. Training and evaluating on such recordings requires either clean reference signals or the use of objectives such as mixture invariant training (MixIT), which does not require clean references. Combining MixIT-style training with vibration conditioning is a technically tractable extension that would constitute a significant step toward real-world deployment.

Extension to more than two sources. Extending the separator output to $K > 2$ sources requires both a higher-capacity separator and a mixture vibration signal that can discriminate among K sources simultaneously. For $K = 3$, the mixture vibration at equal energy carries even less per-source discriminative information than for $K = 2$.

Fixed-mount contact microphone as a real vibration source. A single contact microphone mounted on a fixed structure — such as a tree trunk or monitoring post — would capture structural vibrations dominated by whichever species is vocalizing closest to the mount point, providing genuine per-source discrimination without requiring sensor attachment to any animal.

Larger and more diverse species corpora. Replicating the experiment on a large-scale corpus — Xeno-Canto, BirdCLEF, or a curated multi-habitat PAM archive — would test whether the vibration encoder learns features that are robust across vocalization types ranging from the impulsive calls of woodpeckers to the sustained harmonic songs of warblers.

Simulation fidelity as a controlled variable. A complementary ablation that varies the noise level σ and transient probability p would directly address the sensitivity of the approach to simulation quality, providing a more precise answer to RQ2.

Downstream bioacoustic evaluation. Evaluating whether the separated outputs of Model B, when used as input to a species recogniser such as BirdNET [9], produce higher recognition accuracy than Model A’s outputs would establish a direct

link between the separation metric improvements reported here and the biodiversity monitoring outcomes that motivate this line of research.

7.5 Closing Remark

The woodland recorder with which this thesis opened captures everything faithfully. Whether the analysis pipeline that follows can make use of what it captures depends on the quality of the separation step. This thesis has shown that a computationally simulated vibration signal — one that requires no sensor, no animal handling, and no modification to existing recording infrastructure — can meaningfully improve that separation step, particularly in the high-overlap conditions where automated biodiversity monitoring is most error-prone and ecological interest is greatest.

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